

Article

Improved Sparrow Search Algorithm Applied to Structural Optimization of Transmission Tower

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Abstract

To balance the requirements of structural safety and economic efficiency in transmission tower design, this study proposes a structural optimization method based on an Improved Sparrow Search Algorithm (ISSA). Specifically, the methodology integrates Circle chaotic mapping to initialize the sparrow population distribution, followed by the introduction of a firefly perturbation strategy. This strategy updates the sparrow positions derived from the basic algorithm, thereby enhancing the algorithm's ability to escape local optima. Based on these enhancements, a mathematical model for structural optimization is established with the primary objective of minimizing the total weight of the tower. The penalty function method is employed to handle constraints, while the finite element method is utilized for internal force analysis. To verify the effectiveness of the proposed approach, a typical cat-head transmission tower is selected as a case study. The optimization results obtained from the ISSA are compared with those produced by the Daoheng software, demonstrating a weight reduction of 3.79%. These findings indicate that the ISSA effectively improves the optimization performance and provides an efficient solution for the preliminary sizing design of transmission towers.

Keywords: Improved Sparrow Search Algorithm; transmission tower; finite element analysis; structure optimization

1. Introduction

Currently, China is undergoing a critical phase in the accelerated development of a new type of energy system. Consequently, the construction scale of high-voltage and ultra-high-voltage long-distance power transmission projects is expanding annually. As the most critical supporting structures in power transmission lines, transmission towers primarily withstand various loads from transmission lines and the external environment. Therefore, these structures are characterized by stringent safety requirements and represent a significant portion of the total project expenditure. In light of the “dual carbon” strategy related to carbon peaking and carbon neutrality, optimizing the structure of transmission towers to reduce steel consumption while ensuring structural safety is of significant importance. This approach not only enhances the economic efficiency of engineering projects but also contributes substantially to the reduction of carbon emissions.

Structural optimization is generally categorized into size optimization, shape optimization, and topology optimization [1–3]. The core components of this process involve the



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establishment of a rigorous mathematical model and the selection of a suitable optimization algorithm. A mathematical model typically comprises three fundamental elements: design variables, objective functions, and constraints. The available optimization methodologies are generally categorized into three classes, namely analytical methods, numerical methods, and intelligent optimization algorithms [4]. Analytical methods involve complex solution procedures and are generally unsuitable for large-scale and complex structural optimization problems in practical engineering. Numerical methods for structural topology and sizing optimization problems are primarily classified into two categories, namely mathematical programming methods and optimality criteria methods. Mathematical programming methods are rooted in mathematical optimization theory and function by converting the optimization problem into an extremum problem of a mathematical model, which is subsequently solved using iterative algorithms. Although characterized by high computational accuracy and theoretical rigor, mathematical programming methods often entail a significant computational burden when applied to complex structures, which hinders their direct implementation in large-scale engineering. In contrast, optimality criteria methods are developed by constructing heuristic iterative rules grounded in physical intuition or mathematical optimality conditions. While these methods exhibit superior computational efficiency, they are predominantly limited to addressing optimization problems with simple constraints.

Intelligent optimization algorithms have undergone rapid development in recent years. These algorithms are inspired by natural phenomena or biological behaviors, including genetic evolution, the collective interactions of ant colonies, and the foraging strategies of whales [5–7]. Characterized by strong robustness and high flexibility in diverse applications, these meta-heuristic approaches are frequently employed to address complex structural optimization challenges. Numerous scholars have conducted extensive research on the structural optimization of transmission towers, with their research efforts primarily focusing on the following two dimensions: The first dimension involves enhancing the performance of meta-heuristic algorithms and ensuring their suitability for practical engineering. Given that the structural optimization of transmission towers is characterized by high dimensionality and nonlinearity, researchers frequently integrate biological behavioral patterns observed in nature to augment the efficiency of the optimization process. Scholars such as Tort et al. [8] and Couceiro et al. [9] enhanced the simulated annealing algorithm and integrated it into professional design software to facilitate the rapid iterative calculation of transmission tower structures. Cheng and Shi [10] proposed an improved fruit fly optimization algorithm to address cross-sectional size optimization problems, which significantly enhances search accuracy within complex constraint environments. Baykasoğlu et al. [11] introduced a weighted superposition attraction algorithm based on swarm intelligence to optimize complex truss structures, successfully achieving the minimization of total structural weight. The second dimension focuses on the refinement of mechanical analysis models and the formulation of effective constraint-handling strategies. Since the optimization of transmission towers necessitates the consideration of multiple intricate constraints including structural strength, stability, and stiffness, the fidelity of the mechanical model is of paramount importance because it directly influences the precision and feasibility of the final optimization results. Jahangiri et al. [12] implemented a Prize-Penalty Strategy to manage constraints such as stress and deflection during the optimization of truss structures, which yielded satisfactory results. Furthermore, the supervised classification machine learning method introduced by Rosso et al. [13] demonstrates excellent adaptability to both nonlinear and discontinuous boundaries.

In response to the increasing complexity and expanding scale of practical engineering structures, a growing number of researchers have shifted their focus toward intelligent

optimization algorithms. Specifically, the Sparrow Search Algorithm (SSA) [14] is a swarm intelligence optimization method developed by simulating the natural behaviors of sparrow populations, such as foraging, competition, and anti-predation. The algorithm is characterized by high convergence accuracy and a streamlined parameter configuration, offering significant advantages in addressing optimization problems that are non-linear, multimodal, and high-dimensional [15]. Due to the challenges of non-uniform population distribution and the susceptibility of the SSA to premature convergence toward local optima, this study improves the original algorithm by incorporating Circle chaotic mapping and a firefly perturbation strategy. Building upon these considerations, this study proposes an Improved Sparrow Search Algorithm (ISSA). In conjunction with the finite element analysis of transmission tower structures, the enhanced algorithm is applied to structural optimization with the primary objective of minimizing the total weight of the tower. The results demonstrate that the improved algorithm can be effectively applied to the structural optimization design of transmission towers.

Although Circle chaotic mapping and firefly perturbation are both existing strategies, the focus of this study lies in developing a stage-wise collaborative improvement framework for engineering-constrained optimization problems, tailored to the characteristics of transmission tower structural optimization. Based on the basic SSA, Circle chaotic mapping is introduced to alleviate the problem of uneven initial population distribution, while the firefly perturbation strategy is employed to enhance the local search capability of SSA during the optimization process and improve its ability to escape local optima. These two strategies address the limitations of SSA at different optimization stages. Based on this mechanism, the proposed ISSA is further integrated with finite element analysis, discrete section selection, and stress, slenderness ratio, and displacement constraints of transmission towers, and is applied to the preliminary sizing optimization of primary load-carrying members.

2. Internal Force Analysis of Transmission Towers

The structural optimization of transmission towers is an iterative procedure that begins with the analysis of internal forces, followed by the selection of member cross-sections. This sequence is performed repeatedly until a global optimal solution is identified. Consequently, internal force analysis represents a pivotal stage in the overall optimization design. For the internal force calculations of the transmission tower structures [16,17], the finite element method is employed alongside the robust computational capabilities of MATLAB R2022a to carry out the structural simulations.

2.1. Finite Element Analysis

Finite element analysis is fundamentally categorized into three primary stages, which include discretization, local element analysis, and global structural analysis. For transmission tower structures, discretization involves representing the individual structural members as discrete elements, while the junctions where these members intersect and their connections to the supports are abstracted as nodes. Local element analysis involves the derivation of the mechanical relationship between nodal forces and nodal displacements for each individual member to formulate the element stiffness matrix. Global structural analysis involves aggregating the individual stiffness matrices of all members into a comprehensive global stiffness matrix by utilizing coordinate transformation matrices based on the actual tower configuration. This process subsequently enables the establishment of the global equilibrium equations for the entire system. Prior to the solution of the global equilibrium equations, boundary conditions such as prescribed displacements must be incorporated. This procedure facilitates the determination of critical mechanical parameters

including unknown nodal displacements, element stresses, and strains. In transmission line engineering design, transmission towers are commonly analyzed using spatial truss models, in which primary and bracing members are idealized as two-force elements, and joints are assumed to be pinned. The forces acting on members are equivalently transferred to the nodes. This modeling approach has been widely adopted in current engineering design codes and related engineering software, as it ensures computational efficiency while meeting the accuracy requirements of preliminary design [18–20]. Therefore, in this study, the above truss model is employed in the optimization process to facilitate comparison with commonly used engineering analysis methods. It should be noted that this model does not consider bending effects or semi-rigid joint behavior, and their influence will be further investigated in future work.

2.1.1. Formulation of the Element Stiffness Matrix Equation

Transmission towers belong to spatial truss structures. In structural mechanics, the element local stiffness matrix is transformed into the element stiffness matrix in the global coordinate system through a coordinate transformation matrix. The specific procedure is as follows:

(1) Element stiffness matrix equation in the local coordinate system

The element stiffness matrix in the local coordinate system is shown as follows:

$$\{A_M\}_i = [S_M]_i \{D_M\}_i \quad (1)$$

where $\{A_M\}_i$ is the nodal force vector of member i in the local coordinate system; $[S_M]_i$ is the element stiffness matrix of member element i in the local coordinate system; and $\{D_M\}_i$ is the nodal displacement vector of member i in the local coordinate system.

The element stiffness matrix in the local coordinate system, denoted as $[S_M]_i$, is expressed as:

$$[S_M]_i = \frac{EA}{l} \begin{pmatrix} 1 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (2)$$

where E is the elastic modulus of the material; A is the cross-sectional area of the member; and l is the length of the member.

(2) Element stiffness matrix equation in the global coordinate system

The element stiffness matrix in the global coordinate system is shown as follows:

$$\{A_S\}_i = [S_{MD}]_i \{D_S\}_i \quad (3)$$

where $\{A_S\}_i$ is the nodal force vector of member i in the global coordinate system; $[S_{MD}]_i$ is the element stiffness matrix of member element i in the global coordinate system; and $\{D_S\}_i$ is the nodal displacement vector of member i in the global coordinate system.

Since $\{A_M\}_i$ and $\{A_S\}_i$, as well as $\{D_M\}_i$ and $\{D_S\}_i$, are defined in different coordinate systems, their relationships need to be established using the element coordinate transformation matrix.

The element stiffness matrix in the global coordinate system, denoted as $[S_{MD}]_i$, is expressed as:

$$[S_{MD}] = \frac{EA}{l} \begin{bmatrix} C_x^2 & C_y C_x & C_z C_x & -C_x^2 & -C_y C_x & -C_z C_x \\ C_x C_y & C_y^2 & C_z C_y & -C_x C_y & -C_y^2 & -C_z C_y \\ C_x C_z & C_y C_z & C_z^2 & -C_x C_z & -C_y C_z & -C_z^2 \\ -C_x^2 & -C_y C_x & -C_z C_x & C_x^2 & C_y C_x & C_z C_x \\ -C_x C_y & -C_y^2 & -C_z C_y & C_x C_y & C_y^2 & C_z C_y \\ -C_x C_z & -C_y C_z & -C_z^2 & C_x C_z & C_y C_z & C_z^2 \end{bmatrix} \quad (4)$$

where C_x , C_y , and C_z are the direction cosines of the member axis with respect to the x -, y -, and z -directions of the global coordinate system, respectively, and can be calculated from the coordinates of the two end nodes of the member.

2.1.2. Global Nodal Displacements and Global Nodal Forces

(1) Global nodal displacements of the structure

During the discretization of the transmission tower, the nodes are considered as hinged. Since each node has three independent linear displacements in the x , y and z directions of the global coordinate system, a tower with a total of n nodes will have $3n$ independent linear displacements. The global nodal displacement vector can be expressed as follows:

$$\{D_J\} = \{(D_J)_1 \ (D_J)_2 \ (D_J)_3 \ \cdots \ (D_J)_{3k-2} \ (D_J)_{3k-1} \ (D_J)_{3k} \ \cdots\}^T \quad (5)$$

where $(D_J)_1$, $(D_J)_2$ and $(D_J)_3$ are the linear displacements of node 1 in the x , y and z directions; $(D_J)_{3k-2}$, $(D_J)_{3k-1}$ and $(D_J)_{3k}$ are the linear displacements of node k in the x , y and z directions.

(2) Global nodal forces of the structure

The global nodal forces of the structure are the sum of the total support reactions and the combined nodal loads. Among them, the support reactions are the reaction forces generated at the supports in the direction of the constraints under the action of loads, represented by the vector $\{R_J\}$. The combined nodal loads are represented by the vector $\{A_C\}$. Thus, the global nodal force vector of the structure $\{A_J\}$ is expressed as follows:

$$\{A_J\} = \{R_J\} + \{A_C\} \quad (6)$$

2.1.3. Formulation and Solution of the Global Nodal Stiffness Matrix Equation

According to the numbering of the degree of freedom at the ends of each member, the global element stiffness matrices $[S_{MD}]$ are assembled into the overall stiffness matrix $[S_J]$, thereby establishing the global equilibrium equation of the transmission tower structure:

$$[S_J]\{D_J\} = \{A_J\} \quad (7)$$

In the equation above, the global nodal stiffness matrix $[S_J]$ is singular, which makes the corresponding system of equations unsolvable without additional constraints. Therefore, known displacement boundary conditions, such as the displacements at the connection points between the transmission tower legs and the foundation, must be introduced to solve the global nodal stiffness matrix equation.

2.1.4. Final Member-End Forces

The final member end forces of the structure can be calculated using the following equation:

$$\{A_M\}_i = \{A_{ML}\}_i + [S_M]_i [R_T]_i \{D_J\}_i \quad (8)$$

where $\{A_{ML}\}_i$ is the vector of fixed-end forces generated by the loads acting on the member.

2.2. Internal Force Calculation Process

Following the aforementioned analytical procedure, an internal force analysis program for the transmission tower was developed in MATLAB, the primary workflow of which is illustrated in Figure 1.

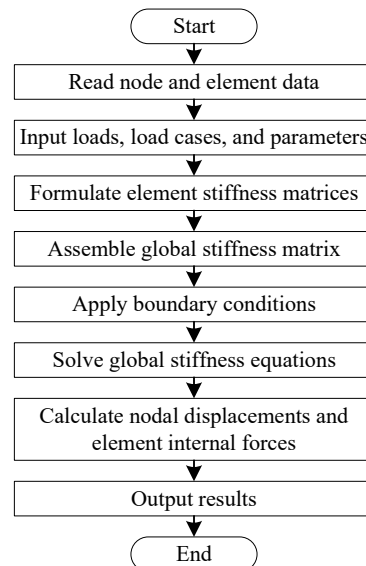


Figure 1. Flow chart of finite element analysis.

2.3. Verification of the MATLAB-Based Finite Element Program

2.3.1. Introduction to the Spatial Structure

To verify the correctness of the MATLAB finite element analysis program in calculating the internal forces of transmission tower structures, a typical three-dimensional lattice tower structure was selected as a verification example. The calculation results obtained from the MATLAB program were compared with those obtained from ANSYS (version 19.2) finite element analysis. The tower structure is shown in Figure 2. Since the structure is symmetric about its central axis, the center of the bottom plane was taken as the coordinate origin to establish a three-dimensional Cartesian coordinate system. The nodal coordinates are listed in Table 1.

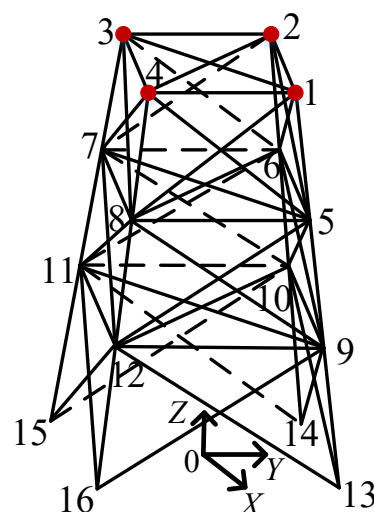


Figure 2. Schematic diagram of a three-dimensional lattice tower structure (numbers indicate node labels).

Table 1. Node coordinates.

Node	X Direction (m)	Y Direction (m)	Z Direction (m)
1	1.50	1.50	10.00
5	1.95	1.95	7.00
9	2.40	2.40	4.00
13	3.00	3.00	0.00

The three-dimensional lattice tower is entirely constructed using Q235 steel, and the member cross-section is $L45 \times 5$. The elastic modulus of the material is $E = 2.06 \times 10^{11}$ Pa. The finite element analysis of the tower structure is carried out using ANSYS, in which the members are modeled with LINK180 elements. The applied loads are shown in Table 2.

Table 2. Applied loads.

Node	X Direction (kN)	Y Direction (kN)	Z Direction (kN)
1	11	40	−20
2	11	40	−20
3	12	50	−20
4	12	50	−20

2.3.2. Result Verification

Under the same nodal coordinates, member connectivity, material parameters, section parameters, loads, and boundary conditions, the spatial tower was analyzed using both the MATLAB program and ANSYS software. The results show that the calculated axial forces of the members obtained by the two methods are basically consistent. The comparison results of selected member forces are shown in Table 3. The selected members cover the upper, middle, and lower parts of the tower, which can reasonably reflect the reliability of the program in calculating member internal forces of spatial tower structures.

Table 3. Comparison of Internal Force Calculation Results.

Member	Location	ANSYS Result (N)	MATLAB Result (N)	Relative Error (%)
1–5	Upper main leg	−51,640	−51,640.5	9.68×10^{-4}
5–9	Middle main leg	−124,900	−124,902.9	2.32×10^{-3}
9–13	Lower main leg	−160,000	−159,997.2	$−1.75 \times 10^{-3}$
1–4	Top horizontal	−1090	−1090.0	0
5–8	Upper-middle horizontal	7874.2	7874.2	0
9–12	Middle horizontal	12,842	12,842.3	2.34×10^{-3}
1–8	Upper diagonal	45,678	45,678.4	8.76×10^{-4}
5–12	Middle diagonal	32,393	32,392.6	$−1.23 \times 10^{-3}$
9–16	Lower diagonal	19,486	19,485.7	$−1.54 \times 10^{-3}$
2–4	Top diaphragm diagonal	−312.38	−312.4	6.40×10^{-3}
6–8	Upper-middle diaphragm diagonal	3700.5	3700.5	0
10–12	Lower-middle diaphragm diagonal	3652.1	3652.1	0

As shown in the table, the axial forces of the selected typical members calculated by the MATLAB program agree well with the ANSYS results, and the maximum absolute relative error is only 0.0064%. Therefore, the MATLAB space truss finite element program established in this study has high reliability in calculating member axial forces and can be used for the subsequent optimization analysis of transmission tower structures.

3. Sparrow Search Algorithm and Its Improvements

3.1. Sparrow Search Algorithm

In the SSA, the sparrow population is divided into producers, scroungers, and sparrows that perceive danger [21,22]. Suppose there are N sparrows in a D -dimensional space; the position of the i -th sparrow can be represented as $X_i = [x_{i1}, \dots, x_{id}, \dots, x_{iD}]$, and the fitness value is $F_x = [f(x_{i1}), \dots, f(x_{id}), \dots, f(x_{iD})]$, where $i \in [1, N]$, $d \in [1, D]$, and both are integers.

In the SSA, producers are individuals with better fitness values in the population and are responsible for searching for promising solution positions in the search space. The position update formula is defined as follows:

$$x_{id}^{t+1} = \begin{cases} x_{id}^t \cdot \exp\left(-\frac{i}{\alpha \cdot iter_{\max}}\right), & R_2 < ST \\ x_{id}^t + Q \cdot L, & R_2 \geq ST \end{cases} \quad (9)$$

where t denotes the current iteration number; $iter_{\max}$ represents the maximum number of iterations; $\alpha \in (0, 1]$ is a random number; Q is a random number following a normal distribution; L represents a $1 \times d$ matrix in which all elements are 1; $R_2 \in [0, 1]$ is the warning value; and $ST \in [0.5, 1]$ denotes the safety value.

When $R_2 < ST$, the environment is safe, and producers conduct a global search. Otherwise, it indicates that the current environment is hazardous, prompting producers to issue an alarm. At this point, all sparrows migrate to other areas.

Individuals other than producers are defined as scroungers, constituting the majority of the population. Their role involves performing a refined local search based on the findings of the producers. The position update formula is expressed as follows:

$$x_{id}^{t+1} = \begin{cases} Q \cdot \exp\left(\frac{x_{\text{worst}} - x_{id}^t}{i^2}\right), & i > \frac{N}{2} \\ x_p^{t+1} + |x_{id}^t - x_p^{t+1}| \cdot A^+ \cdot L, & i \leq \frac{N}{2} \end{cases} \quad (10)$$

where x_{worst} denotes the global worst position of the population; x_p represents the optimal position of the producers; A is a $1 \times d$ matrix with elements randomly assigned values of 1 or -1 , where $A^+ = A^T(AA^T)^{-1}$.

When $i > N/2$, the follower is in a starving state and will randomly migrate to other locations for foraging. Otherwise, the individual moves toward the producers currently occupying the optimal position to forage.

Furthermore, 10% to 20% of the population consists of sparrows that perceive danger, which monitor environmental risks via random walks and issue alarms. The position update formula is defined as follows:

$$x_{id}^{t+1} = \begin{cases} x_{\text{best}}^t + \beta \cdot |x_{id}^t - x_{\text{best}}^t|, & f_i > f_g \\ x_{id}^t + K \cdot \left(\frac{|x_{id}^t - x_{\text{worst}}^t|}{(f_i - f_w) + \varepsilon}\right), & f_i = f_g \end{cases} \quad (11)$$

where x_{best}^t denotes the optimal position within the population during the t -th iteration; β is the step size control parameter, with $\beta \sim N(0, 1)$; $K \in [-1, 1]$ represents a random number; f_i is the fitness value of the i -th sparrow; f_w and f_g are the worst and best fitness values of the population, respectively; and ε is an infinitesimal constant.

When $f_i > f_g$, the individual is at the periphery of the population and faces a risk of predation. When $f_i = f_g$, the individual is located at the center of the population and

senses danger, leading it to actively move closer to other members of the population to reduce the risk of being preyed upon.

3.2. Circle Chaotic Map

To address issues such as local optima entrapment and stagnant convergence when performing structural optimization with the basic SSA, the Circle chaotic map is introduced. As a discrete dynamical system with strong chaotic characteristics, its core advantages lie in high coverage and excellent robustness. By leveraging the ergodic properties and stochastic patterns of chaotic systems, the diversity and spatial distribution uniformity of the initial population are enhanced [23–25], which effectively accelerates the convergence rate. The mathematical model is expressed as follows:

$$x_{i+1} = \text{mod} \left(x_i + 0.2 - \left(\frac{0.5}{2\pi} \right) \sin(2\pi x_i), 1 \right) \quad (12)$$

where $i + 1$ denotes the dimensionality of the solution. To provide an intuitive illustration of the Circle chaotic map, its distribution plot and histogram are generated to evaluate whether the sample points are uniformly distributed. To ensure a sufficient sample size and to more accurately demonstrate the distribution characteristics of the chaotic values within the interval, 1000 sample points are selected. The distribution plot and histogram for the initial solutions generated by the Circle chaotic map are presented in Figure 3.

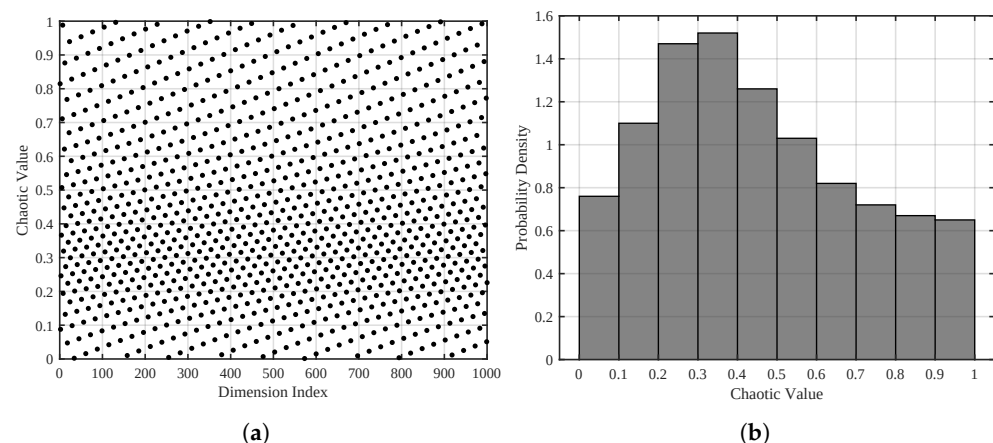


Figure 3. Circle chaotic map results. (a) Distribution plot. (b) Histogram.

3.3. Firefly Perturbation Strategy

The Firefly Algorithm [26] is a swarm intelligence optimization algorithm based on the behavioral patterns of firefly populations. The core concept of the algorithm involves facilitating global search through a mechanism of brightness comparison and distance-based attraction among fireflies. Integrating the FA into the traditional SSA enhances the capacity to escape local optima and improves search precision. This approach is typically applied to address nonlinear problems and combinatorial optimization tasks.

To address the tendency of the SSA to fall into local optima during structural optimization, a firefly perturbation mechanism is introduced to enhance the local exploration capability. By simulating the mutual attraction between fireflies, directional perturbations are applied to individual positions. The fitness values are compared before and after the perturbation, and the superior one is selected to update the individual status, thereby enhancing population diversity and improving the global optimization capability [27].

Specifically, the attractiveness between fireflies serves as the core regulatory parameter for perturbation intensity, and its formula is defined as follows:

$$\beta = \beta_0 \cdot e^{-\gamma r_{ij}^2} \quad (13)$$

where β_0 denotes the maximum attractiveness; γ represents the light absorption coefficient; and r_{ij} is the distance between fireflies i and j .

Between fireflies i and j , if the brightness of j is superior to that of i , firefly i will be attracted to j and its position will be updated. The position update is described as follows:

$$x_i = x_i + \beta \cdot (x_j - x_i) + \alpha \cdot (\text{rand} - 0.5) \quad (14)$$

where x_i and x_j denote the spatial positions of fireflies i and j , respectively; and $\alpha \in [0, 1]$ represents the step size factor.

4. Member Cross-Section Optimization Model and Process

4.1. Member Cross-Section Optimization Model

The preceding sections primarily discussed the methods employed for the structural optimization of transmission towers. However, achieving an optimal design for such structures requires not only the selection of appropriate and efficient optimization methods but also the establishment of a reliable mathematical model. For the structural optimization of transmission towers, a mathematical model is established with the objective of minimizing the total weight of the tower, subject to constraints such as member stress, slenderness ratio, and nodal displacement.

Regarding the structural optimization design of transmission towers, the primary constraints considered are as follows:

(1) Member stress

$$\sigma_i = \begin{cases} \frac{N_i^+}{A_i} \leq f, & N_i^+ \text{ is the tensile internal force} \\ \frac{N_i^-}{\varphi_i A_i} \leq f, & N_i^- \text{ is the compressive internal force} \end{cases} \quad (15)$$

where σ_i denotes the stress of the i -th member; f represents the design value of the steel strength; and φ_i is the stability coefficient of the axially compressed member.

(2) Slenderness ratio

$$\begin{cases} \lambda_i - [\lambda]_C \leq 0 \\ \lambda_i - [\lambda]_T \leq 0 \end{cases} \quad (16)$$

where λ_i denotes the slenderness ratio of the i -th member, and $[\lambda]_C$ and $[\lambda]_T$ represent the allowable slenderness ratios for members under compression and tension, respectively.

(3) Nodal displacement

$$u_j - [u] \leq 0 \quad (17)$$

where u_j denotes the displacement of the j -th node, and $[u]$ represents the allowable displacement.

In this study, stability constraints based on the slenderness ratio are adopted as the criterion for evaluating member stability. This approach is derived from current transmission tower design codes and is widely used in the preliminary design stage of engineering practice. It should be noted that geometric nonlinear effects, global buckling, and second-order effects are typically considered in the detailed design stage, rather than in preliminary sizing optimization. Therefore, geometric nonlinear analysis and buckling

analysis are not included in this study, in order to improve computational efficiency and ensure comparability with engineering design practices. The above simplifications may not fully capture complex stability behavior, and their influence will be further investigated in future work.

With the objective of minimizing the structural weight of the transmission tower, the objective function is formulated as follows:

$$W = \sum_{i=1}^n \rho_i l_i A_i \quad (18)$$

where W denotes the total weight of the transmission tower; $A_i \in S$ represents the cross-sectional area of each member within the tower; $S = \{S_1, S_2, \dots, S_{NS}\}$ is the discrete set of available member cross-sections; ρ_i is the material density of the i -th member; l_i is the element length of the i -th member; and n is the total number of members in the tower structure.

Based on the analysis above, the structural optimization of transmission towers exhibits significant constraint characteristics. These constraints span multiple dimensions, including strength, stiffness, stability, and detailing requirements [28], which significantly restrict the feasible region and render the search space complex, high-dimensional, and non-linear, making the optimization process highly challenging. To address such problems, methods such as the penalty function method, feasibility rules, and stochastic ranking methods are commonly employed. Among these, the penalty function method has become one of the most widely applied constraint handling techniques in tower optimization and similar engineering problems due to its straightforward principle, ease of implementation, and high adaptability to complex constraints [29,30]. The principle of this method can be briefly explained as applying specialized weighted processing to the constraints and then integrating them with the objective function to generate a new objective function. By introducing penalty functions to handle constraints related to member stress, slenderness ratio, and nodal displacement, the problem is effectively transformed into an unconstrained optimization task. Based on this approach, the objective function expression is established as follows:

$$W = \sum_{i=1}^n \rho_i l_i A_i \left(1 + \sum_{i=1}^n \max \left[0, \left(\frac{\sigma_i}{f} - 1 \right) \right] + \sum_{i=1}^n \max \left[0, \left(\frac{\lambda_i}{[\lambda]} - 1 \right) \right] + \sum_{j=1}^m \max \left[0, \left(\frac{u_j}{[u]} - 1 \right) \right] \right)^2 \quad (19)$$

To handle the stress, slenderness ratio, and nodal displacement constraints, a normalized quadratic penalty function is adopted in this study. Different types of constraints are first transformed into dimensionless relative violation terms, so that constraint quantities with different physical units can be evaluated in a unified manner. The penalty term is constructed based on the square of the accumulated positive constraint violations. For a feasible solution, all constraint violation terms are equal to zero, and the penalized objective function is identical to the structural weight. For an infeasible solution, the objective function increases with the square of the overall violation degree, thereby reducing the competitiveness of infeasible individuals in the fitness evaluation. This formulation strengthens the penalty effect when multiple constraints are violated simultaneously, while avoiding the introduction of additional manually adjusted penalty coefficients. During each iteration, the structural responses, including member stresses, slenderness ratios, and nodal displacements, are calculated based on the finite element analysis results, and the corresponding constraint satisfaction is then checked. In addition, feasible solutions are

preferentially retained when comparing candidate solutions with comparable objective values, which helps guide the population toward the feasible region.

4.2. Optimization Process

Based on the discussion above, when applying the basic SSA to the structural optimization of transmission towers, improvements are introduced via the Circle chaotic map and the Firefly Algorithm. The main optimization process is illustrated in Figure 4.

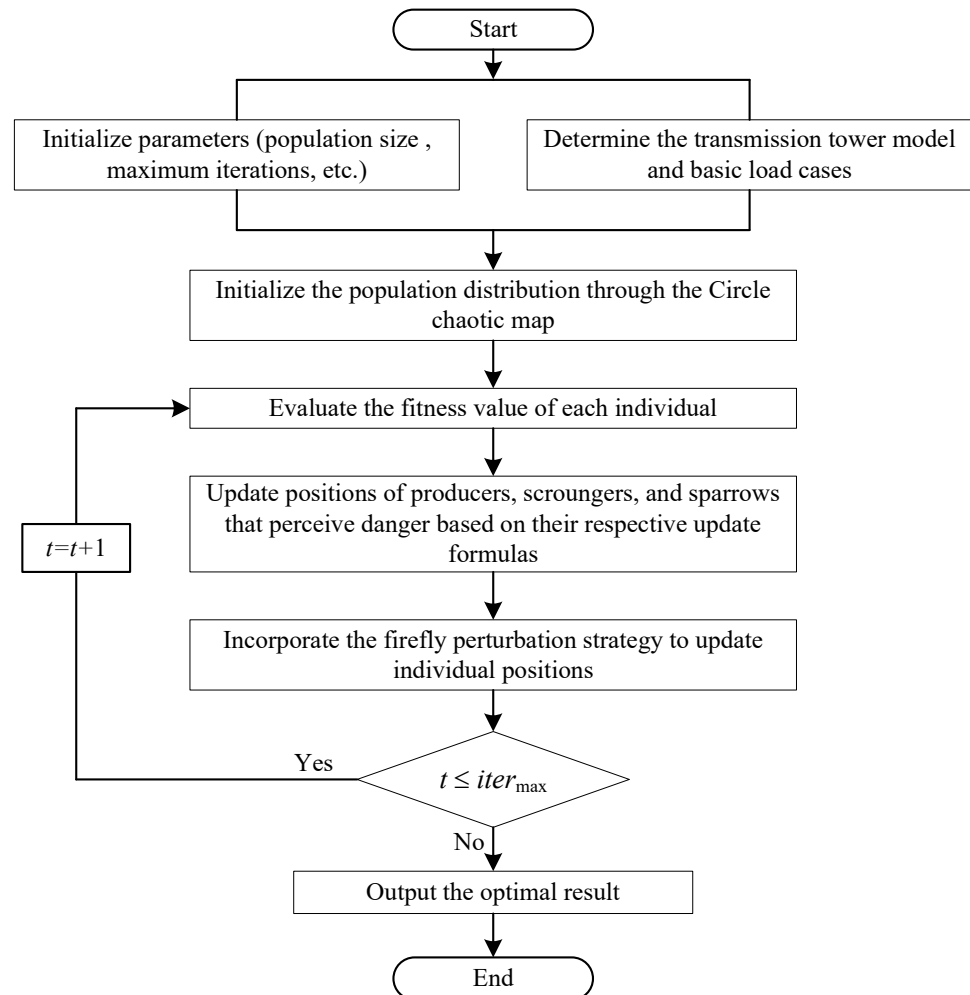


Figure 4. Size optimization design process.

5. Algorithm Verification

5.1. Structure of the Spatial Truss

To compare ISSA with other algorithms, a typical 25-bar spatial truss structure is selected as the optimization object. This structure exhibits spatial truss characteristics and shares similarities with transmission tower structures in terms of force transmission and spatial member configuration. The structural configuration is illustrated in Figure 5.

A 25-member spatial truss structure consisting of 10 nodes is considered. Owing to the symmetry of the structure about the central axis, the center of the base is taken as the origin of the coordinate system. Partial nodal coordinates are listed in Table 4. The material properties are specified as Young's modulus $E = 6.895 \times 10^{10}$ Pa and density $\rho = 2768.8$ kg/m³. The structural constraints are defined as follows: the allowable stress of members is 2.759×10^8 Pa, and the allowable nodal displacements in the x , y , and z directions are 0.889 cm. The 25 members are divided into 8 groups, as detailed in Table 5,

and the loading conditions are given in Table 6. Specifically, the top horizontal member 1–2 has a length of 190.5 cm and forms an angle of 45° with the x -axis in the horizontal plane. The z -axis passes through the midpoint of member 1–2; therefore, the x - and y -coordinates of nodes 1 and 2 can be expressed as $\pm 95.25/\sqrt{2}$ cm.

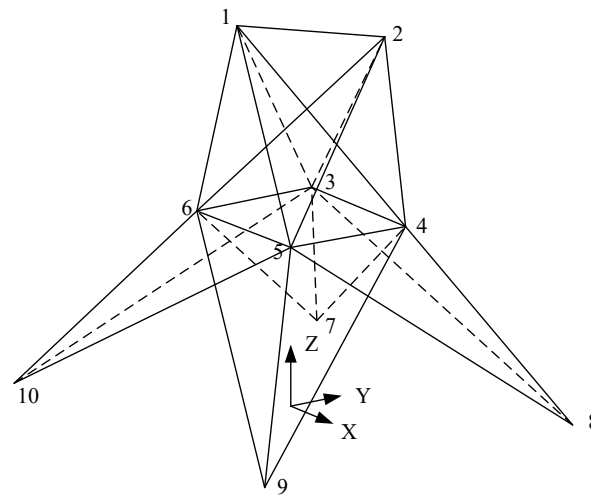


Figure 5. 25-bar spatial truss structure (numbers indicate node labels).

Table 4. Nodal coordinates of the 25-bar spatial truss structure.

Node	X Direction (cm)	Y Direction (cm)	Z Direction (cm)
1	$-95.25/\sqrt{2}$	$-95.25/\sqrt{2}$	508
2	$95.25/\sqrt{2}$	$95.25/\sqrt{2}$	508
4	95.25	95.25	254.00
8	254.00	254.00	0.00

Table 5. Member connectivity and grouping of the structure.

Group No.	Members	Group No.	Members
1	1–2	5	3–4, 5–6
2	1–4, 2–3, 1–5, 2–6	6	3–10, 6–7, 4–9, 5–8
3	2–4, 2–5, 1–3, 1–6	7	3–8, 4–7, 6–9, 5–10
4	3–6, 4–5	8	3–7, 4–8, 5–9, 6–10

Table 6. Loading conditions.

Node	P_x (kN)	P_y (kN)	P_z (kN)
1	4.45	−44.5	−44.5
2	0	−44.5	−44.5
3	2.225	0	0
6	2.67	0	0

The discrete set of cross-sectional areas available for members is defined as $S = \{0.6452, 1.2904, 1.9356, 2.5808, 3.2260, 3.8712, 4.5164, 5.1616, 5.8068, 6.4520, 7.0972, 7.7424, 8.3876, 9.0328, 9.6780, 10.3232, 10.9684, 11.6136, 12.2588, 12.9040, 13.5492, 14.1944, 14.8396, 15.4848, 16.1300, 16.7752, 17.4204, 18.0656, 18.7108, 19.3560, 20.0012, 20.6464, 21.2916, 21.9368\}$ cm². Both SSA and ISSA are applied to optimize the cross-sectional design of the structure. The algorithm parameters are set as follows: population size $N = 50$; explorers account for 20% of the population, while the remaining individuals are followers; warning individuals

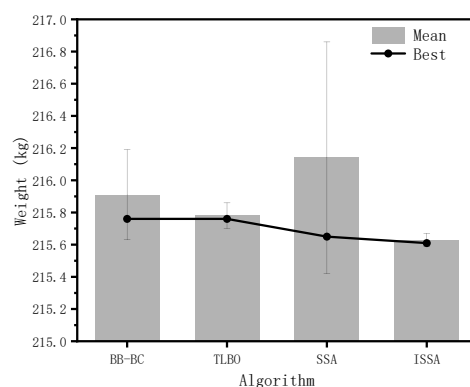
account for 20%; the maximum number of iterations is $\text{iter}_{\max} = 200$, and $ST = 0.8$. For the firefly perturbation strategy, the parameters are set as $\beta_0 = 1$, $\gamma = 1$, and $\alpha = 0.2$.

5.2. Comparison of Results

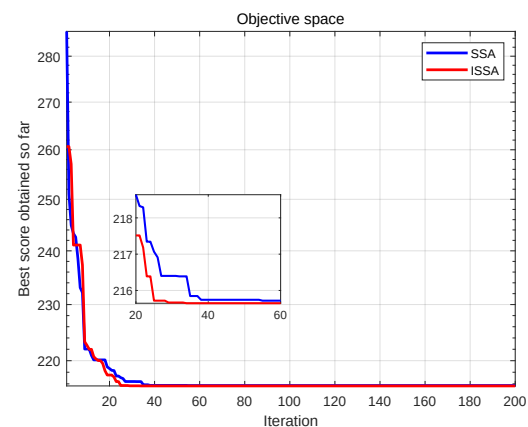
Both SSA and ISSA are independently executed 20 times. Their results are compared with those of the Big Bang–Big Crunch (BB–BC) algorithm [31] and the Teaching–Learning–Based Optimization (TLBO) algorithm [32]. The comparison results are presented in Table 7 and Figure 6a, where the error bars indicate standard deviations. The convergence curves of SSA and ISSA are shown in Figure 6b.

Table 7. Comparison of optimization results.

	BB–BC	TLBO	SSA	ISSA
Best value (kg)	215.76	215.76	215.65	215.61
Mean value (kg)	215.91	215.78	216.14	215.63
Standard deviation (kg)	0.28	0.08	0.72	0.04



(a)



(b)

Figure 6. Comparison of algorithms. (a) Comparison of optimization results. (b) Comparison of convergence curves.

As shown in Table 7 and Figure 6a, ISSA achieves the lowest optimal and average structural weights compared to the other algorithms, and its standard deviation is significantly smaller than those of the other three methods. In particular, compared with SSA, although the optimal values are similar, ISSA obtains lower average values and smaller standard deviations due to the introduction of Circle chaotic mapping and firefly perturbation strategies. Furthermore, as illustrated in Figure 6b, ISSA converges to a stable solution within fewer iterations, indicating a faster convergence speed and stronger stability. These results demonstrate that ISSA is effective for the optimization design of spatial truss structures and can improve the overall optimization performance.

6. Engineering Application

6.1. Overview of the Transmission Tower

The 220 kV cat-head transmission tower model 2A5-ZM1-1 is selected as the object for structural optimization design. This tower is a single-circuit spatial truss system with a nominal height of 18 m and a base width of 4.605 m. It is designed for altitudes below 1000 m and Category B ground roughness. The environmental parameters include a design wind speed of 29 m/s and an ice thickness of 10 mm. The conductor and ground wire are $2 \times \text{LGJ-300/40}$ and JLB-150, respectively. The structural members consist of angle steel made of Q235 and Q345 steel, with an elastic modulus of $E = 2 \times 10^{11}$ Pa and a mass

density of $\rho = 7850 \text{ kg/m}^3$. The meteorological and design conditions for this transmission tower are presented in Table 8 and Table 9, respectively.

Table 8. Meteorological conditions of the tower.

Condition	Temperature (°C)	Wind Speed (m/s)	Ice Thickness (mm)
Basic wind speed	−5	29	0
Icing	−5	10	10
Maximum temperature	40	0	0
Minimum temperature	−40	0	0
Broken conductor	−5	0	10
Installation condition	−15	10	0
Uneven icing	−5	10	10
Annual average temperature	5	0	0

Table 9. Design conditions of iron tower.

Design Condition	Horizontal Span (m)	Vertical Span (m)	Ruling Span (m)	Angle Degree (°)
Value	350	450	400	0

In modeling the transmission tower structure, the members are idealized as bar elements, and the connections between members are assumed to be ideal pinned joints. Under the loading conditions considered in this study, the structural behavior is dominated by axial forces, while torsional effects are relatively weak. Horizontal diaphragms mainly provide torsional stiffness, and their contribution to the overall stiffness is limited. Moreover, their inclusion in the finite element model may introduce redundant constraints and lead to numerical singularities; therefore, they are simplified in this study. Auxiliary members primarily serve to provide lateral support for main and bracing members, and their effects are usually accounted for indirectly through effective length or stability coefficients. In engineering analysis, they are generally not explicitly included in the global structural calculation, and thus are not explicitly modeled in this study. These simplifications may have some influence on local stiffness and stability; however, their impact on the overall optimization trends for the preliminary sizing of primary load-carrying members is limited. The resulting simplified model contains 570 members and 171 nodes; the detailed dimensions are illustrated in Figure 7.

The structural optimization of the transmission tower is conducted with a typical high wind load condition as the primary scenario. In this case, the wind speed is $v = 29 \text{ m/s}$, the temperature is $t = -5^\circ\text{C}$, and there is no ice coverage. The loads primarily consist of wind load and tower self-weight. Specifically, the formula for calculating the magnitude of the wind load on the conductors and ground wires of the transmission tower is as follows:

$$W_x = \beta_c \cdot \alpha_L \cdot W_0 \cdot \mu_Z \cdot \mu_{SC} \cdot d \cdot L_P \cdot B \cdot \sin^2 \theta \quad (20)$$

where W_x denotes the characteristic wind load perpendicular to the conductor and ground wire (kN); β_c represents the gust factor for the conductor and ground wire; α_L is the span reduction factor; W_0 denotes the reference wind pressure (kN/m^2); μ_Z represents the height variation factor of wind pressure; μ_{SC} is the shape coefficient for the conductor and ground wire, where values of 1.0 and 1.1 are adopted for the conductor and ground wire, respectively; d is the outer diameter of the conductor and ground wire (m); L_P denotes the horizontal span of the tower; B is the enlargement coefficient for wind load under icing

conditions; and θ is the angle between the wind direction and the conductor or ground wire ($^{\circ}$).

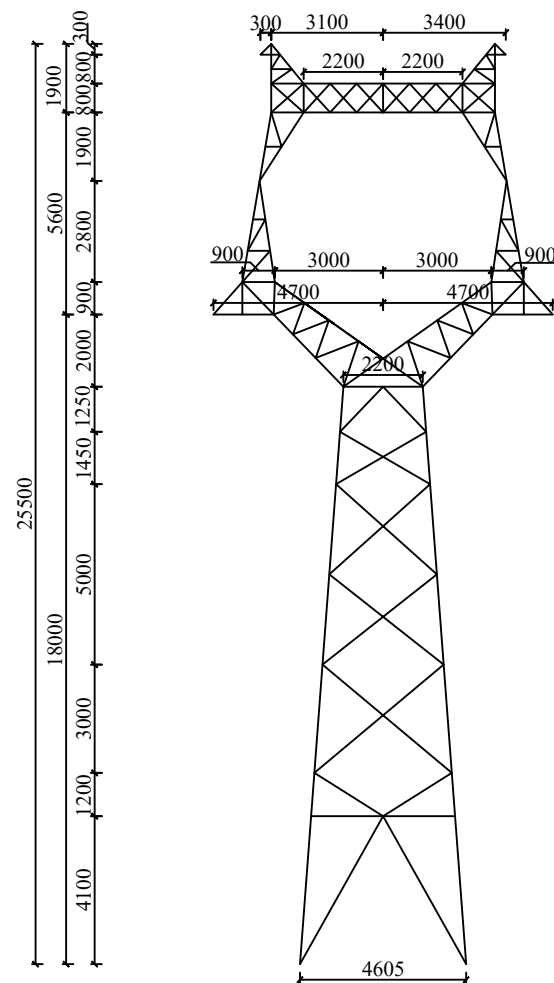


Figure 7. Transmission tower model (mm).

The transmission tower is segmented to calculate the wind load for each section, and the calculated values are then applied to the primary nodes on the windward side. For example, the tower body can be divided into an upper segment and a lower segment with heights of 7.7 m and 4.2 m, respectively. The primary calculation formula is as follows:

$$W_s = W_0 \cdot \mu_z \cdot \mu_s \cdot \beta_z \cdot B \cdot A_s \quad (21)$$

where W_s denotes the characteristic wind load on the transmission tower (kN); μ_s represents the shape coefficient of the members; β_z is the wind vibration coefficient of the tower at height z ; and A_s is the projected area of the members on the windward side (m^2).

6.2. Structural Optimization Analysis of the Transmission Tower

Under identical conditions, the ISSA and Daoheng software(V7.0) are respectively employed to optimize the member sizing for the transmission tower, and the results of both methods are compared.

1. Size Optimization of Transmission Towers Based on the ISSA

The ISSA is applied to the optimal design of transmission towers, with the minimum tower weight taken as the objective function. Here, the tower weight only includes the

weight of angle steel. The parameters involved in the constraints are set according to the Standard for Design of Steel Structures GB 50017-2017 [33], as follows:

(1) Slenderness ratio constraint of members

Regarding the specification of the allowable slenderness ratio $[\lambda]$ for transmission tower members, the maximum allowable slenderness ratios for compressed main members, other compressed members, and tension members are 150, 200, and 400, respectively.

(2) Member stress constraint

In the member stress constraints, the selection of the stability coefficient φ for axially compressed members must be determined based on the slenderness ratio λ of the members. The primary formulas are as follows:

$$\text{When } \lambda_n = \frac{\lambda}{\pi} \sqrt{\frac{f_y}{E}} \leq 0.215: \quad \varphi = 1 - \alpha_1 \lambda_n^2 \quad (22)$$

When $\lambda_n > 0.215$:

$$\varphi = \frac{1}{2\lambda_n^2} \left[(\alpha_2 + \alpha_3 \lambda_n + \lambda_n^2) - \sqrt{(\alpha_2 + \alpha_3 \lambda_n + \lambda_n^2)^2 - 4\lambda_n^2} \right] \quad (23)$$

where f_y is the yield strength of the steel, which is taken as 235 N/mm² for Q235 steel and 345 N/mm² for Q345 steel; E represents the elastic modulus of the material; and α_1 , α_2 , and α_3 are coefficients determined based on different cross-sectional categories. For Class *a* sections, these values are 0.41, 0.986, and 0.152, respectively; and for Class *b* sections, these values are 0.65, 0.965, and 0.3, respectively.

The design strength f of the structural steel for the transmission tower is presented in Table 10.

Table 10. Design values of steel strength.

Material	Thickness or Diameter (mm)	f (N/mm ²)
Q235 steel	≤ 16	215
	$> 16-40$	205
Q345 steel	≤ 16	305
	$> 16-40$	295

(3) The nodal displacement constraints

The selected transmission tower is a self-supporting suspension tangent tower, and according to the design specifications, its deflection limit is defined as $3h/1000$, where h represents the height from the tower foundation to the specific calculation point.

The structural members of the transmission tower are categorized into 17 groups, and the selected angle steel specifications are restricted to no less than L40×3. The discrete set S of available cross-sections for the members is established as $S = \{L40 \times 3 (2.36 \text{ cm}^2), L40 \times 4 (3.09 \text{ cm}^2), L45 \times 4 (3.49 \text{ cm}^2), L50 \times 4 (3.9 \text{ cm}^2), L50 \times 5 (4.8 \text{ cm}^2), L56 \times 4 (4.39 \text{ cm}^2), L56 \times 5 (5.42 \text{ cm}^2), L63 \times 5 (6.14 \text{ cm}^2), L70 \times 5 (6.88 \text{ cm}^2), L75 \times 5 (7.41 \text{ cm}^2), L75 \times 6 (8.8 \text{ cm}^2), L80 \times 6 (9.4 \text{ cm}^2), L80 \times 7 (10.86 \text{ cm}^2), L90 \times 7 (12.3 \text{ cm}^2), L90 \times 8 (13.94 \text{ cm}^2), L100 \times 7 (13.80 \text{ cm}^2), L100 \times 8 (15.64 \text{ cm}^2), L110 \times 8 (17.24 \text{ cm}^2), L110 \times 10 (21.26 \text{ cm}^2), L125 \times 12 (28.91 \text{ cm}^2), L140 \times 12 (32.51 \text{ cm}^2), L160 \times 10 (31.50 \text{ cm}^2)\}$, encompassing a total of 22 types of angle steel profiles.

In summary, the structural optimization of the transmission tower is performed under high wind load conditions using the ISSA, with specific parameter values set as follows: in the SSA, the population size is $N = 100$, where producers and sparrows that perceive

danger each account for 20% of the population, with the remainder as scroungers; the maximum number of iterations $iter_{max}$ is 500, and the safety threshold ST is 0.8; and in the Firefly Algorithm, the parameters are $\beta_0 = 1$, $\gamma = 1$, and $\alpha = 0.2$. The selected member profiles as well as the optimal tower weight after optimization are presented in Table 11, and the convergence curve during the optimization process is illustrated in Figure 8.

Table 11. The optimized transmission tower bar model.

Group No.	Angle Steel Profile	Material
1	L40 × 3	Q235
2	L40 × 4	
3	L45 × 4	
4	L50 × 4	
5	L50 × 5	
6	L63 × 5	
7	L56 × 4	
8	L63 × 5	
9	L45 × 4	
10	L56 × 5	
11	L63 × 5	Q345
12	L70 × 5	
13	L75 × 5	
14	L75 × 6	
15	L100 × 7	
16	L160 × 10	
17	L140 × 12	
Optimal weight (kg)		3872.52

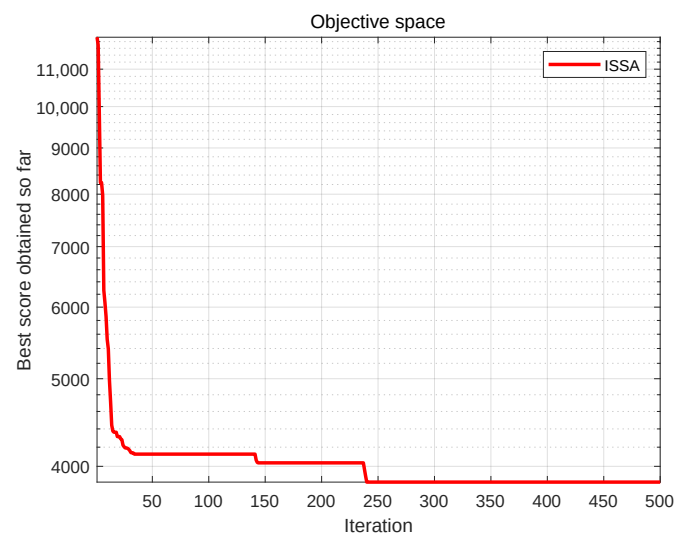


Figure 8. Convergence curve of transmission tower cross-section optimization.

As shown in the table, the final weight of the transmission tower obtained by applying the ISSA to the structural optimization design is 3872.52 kg. When the optimal value is achieved, the regions with higher stress values among the tower members occur in the main members at the bottom of the upper cross-arm, the outer main members below the upper curved arm of the tower head, and the diagonal members in the middle of the tower body; the stress values are 198 N/mm², 143 N/mm², and 200 N/mm², respectively, all of which remain below the corresponding allowable stresses. Furthermore, the higher slenderness ratios among the tower members appear in the main members of the lower

cross-arm, the side diagonal members of the second tower body segment, the diagonal members in the middle of the tower body, and the diagonal members of the tower legs; the values are 140, 189, 187, and 173, respectively, which are all within the allowable maximum slenderness ratios. This indicates that the ISSA optimization is performed effectively while ensuring the structural stability of the tower.

2. Material Selection Optimization of the Transmission Tower Based on Daoheng Software

Under identical conditions, the material selection for the transmission tower is optimized using Daoheng software through fully stressed analysis. The procedure first requires defining the hanging points for the conductors and ground wires of the tower. Subsequently, load calculations are performed within the system, which involves selecting parameters such as the terrain category and surface roughness, tower type and voltage level, meteorological conditions, conductor and ground wire specifications, horizontal span, representative span, and relevant load cases. Finally, the calculated loads are imported into the fully stressed module to obtain the material selection results; the principal workflow is illustrated in Figure 9.

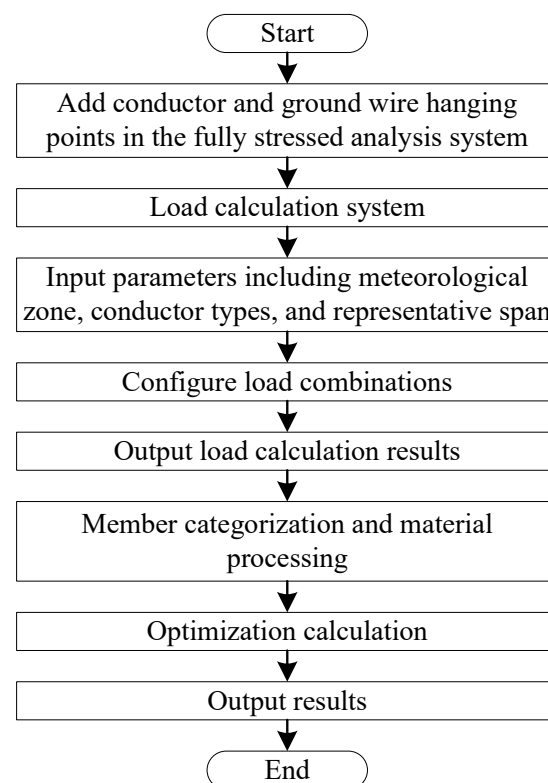


Figure 9. Fully stressed analysis process.

Ultimately, the total weight of the transmission tower obtained through material selection optimization using Daoheng software is 4025.2 kg. Specifically, the main members and diagonal members weigh 1690.3 kg and 2334.8 kg, respectively; meanwhile, Q235 steel and Q345 steel account for 65% and 35% of the total material composition, respectively.

3. Comparative Analysis of Optimization Results

Based on the optimization results from the ISSA and Daoheng software, the optimal tower weights achieved by the two methods are 3872.52 kg and 4025.2 kg, respectively. A comparison of the specific member profiles selected after the optimization process is provided in Table 12. By synthesizing the data from Tables 11 and 12, the comparative analysis indicates that the ISSA selected lighter sections for specific members, leading

to a total weight reduction of 152.68 kg, which corresponds to a weight reduction of approximately 3.79%. Consequently, applying the ISSA to transmission tower optimization holds significant practical engineering importance. Although the weight reduction ratio of a single transmission tower is not very significant, transmission lines are generally characterized by large-scale and repetitive construction, and a large number of towers of the same type are often used. Therefore, the weight reduction of a single tower can lead to cumulative material savings in engineering applications. Reducing the tower weight can not only decrease steel consumption, but may also reduce costs associated with galvanizing, transportation, lifting, and installation. In addition, a lighter tower can reduce the vertical load transferred to the foundation to some extent, thereby providing certain favorable conditions for foundation design and construction. It should be noted that the actual economic benefits still depend on factors such as steel price, fabrication cost, transportation distance, and construction conditions.

Table 12. Comparison of optimized material selections for selected members using the two methods.

Member	Daoheng Software	ISSA
Top ear diagonal members	$L40 \times 3$	$L40 \times 3$
Upper cross-arm main members	$L40 \times 4$	$L40 \times 4$
Upper cross-arm front diagonals	$L40 \times 3$	$L40 \times 3$
...	...	...
Outer main members of the upper curved arm	$L45 \times 5$	$L45 \times 4$
Members at the left and right conductor hanging points	$L70 \times 5$	$L70 \times 5$
Diagonal members of the lower curved arm	$L40 \times 3$	$L40 \times 4$
Diagonal members at the intersection of the lower curved arm	$L70 \times 5$	$L63 \times 5$
...	...	...
Internal diagonal members of the upper horizontal diaphragm	$L40 \times 3$	$L40 \times 3$
Front and back diagonal members of the upper horizontal diaphragm	$L56 \times 5$	$L56 \times 4$
...	...	...
Main members of the uppermost segment of the tower body	$L110 \times 7$	$L100 \times 7$
Front and back diagonal members of the second tower body segment	$L45 \times 4$	$L45 \times 4$
...	...	...
Main members of the fifth segment of the tower body	$L140 \times 12$	$L140 \times 12$
Main members of the lowermost segment of the tower body	$L140 \times 12$	$L160 \times 10$
...	...	...
Diagonal members of the lower horizontal diaphragm	$L56 \times 4$	$L56 \times 4$
Diagonal members at the tower legs	$L160 \times 10$	$L160 \times 10$
Main members of the tower legs	$L140 \times 12$	$L140 \times 12$

Note: "... " indicates omitted values.

Although this study uses only a 220 kV cat-head transmission tower as the engineering example, the proposed ISSA-based optimization framework is not limited to this tower type. Since the framework consists of finite element analysis, discrete member section selection, and constraint evaluation based on stress, slenderness ratio, and nodal displacement, it can

be extended to other lattice transmission towers if the corresponding geometry, load cases, member grouping, section database, and design constraints are properly defined. However, further case studies and detailed design verification are still required for tower structures with complex auxiliary members, semi-rigid joints, or significant bending effects.

7. Conclusions

To address the limitations of the SSA in the discrete section optimization of transmission towers, this study introduces Circle chaotic mapping and a firefly perturbation strategy to improve the algorithm. By integrating the improved algorithm with the finite element analysis method, an optimization model for the primary load-carrying members of transmission towers is established. It should be noted that the structural modeling approach and associated simplifications adopted in this study are mainly intended for the preliminary sizing optimization of primary members, and the results are primarily used to evaluate the performance of different optimization algorithms. For practical engineering applications, further verification is still required at the detailed design stage by employing refined finite element models, multiple loading conditions, and nonlinear analysis methods to ensure structural safety and reliability. The main conclusions are as follows:

- (1) To address the deficiencies of SSA at different stages in the discrete section optimization of transmission towers, Circle chaotic mapping and the firefly perturbation strategy are introduced, thereby enhancing both the global exploration and local exploitation capabilities of SSA in transmission tower optimization problems.
- (2) A 220 kV cat-head transmission tower is taken as an engineering example. By combining a MATLAB-based finite element analysis program with ISSA, preliminary sizing optimization of the primary load-carrying members is performed. The results show that, under the constraints of member strength, stability, and nodal displacement, ISSA can obtain a lighter structural scheme, demonstrating its effectiveness for the preliminary sizing optimization of primary members in the studied tower type.
- (3) Compared with the design results obtained from Daoheng software, the ISSA-optimized scheme achieves a weight reduction of 152.68 kg, corresponding to a reduction ratio of approximately 3.79%. Although the reduction ratio for a single tower is moderate, considering the repetitive construction characteristics of transmission lines, this result indicates certain practical value in terms of material savings and engineering applications.
- (4) This study is conducted based on a single tower type and a typical controlling load case, and some structural components are simplified in the model. Therefore, the applicability of the proposed method to other tower types, complex load combinations, and refined engineering models still requires further investigation in future work.

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